

# Towards Practical Scene Change Detection: A Fast, Unaligned Video Framework via Spatiotemporal Alignment

Yi-Heng Yeh<sup>1</sup>, Chuan-Yuan Huang<sup>1</sup>, Kuan-Wen Chen<sup>1\*</sup>, and Li-Yu Lu<sup>2</sup>

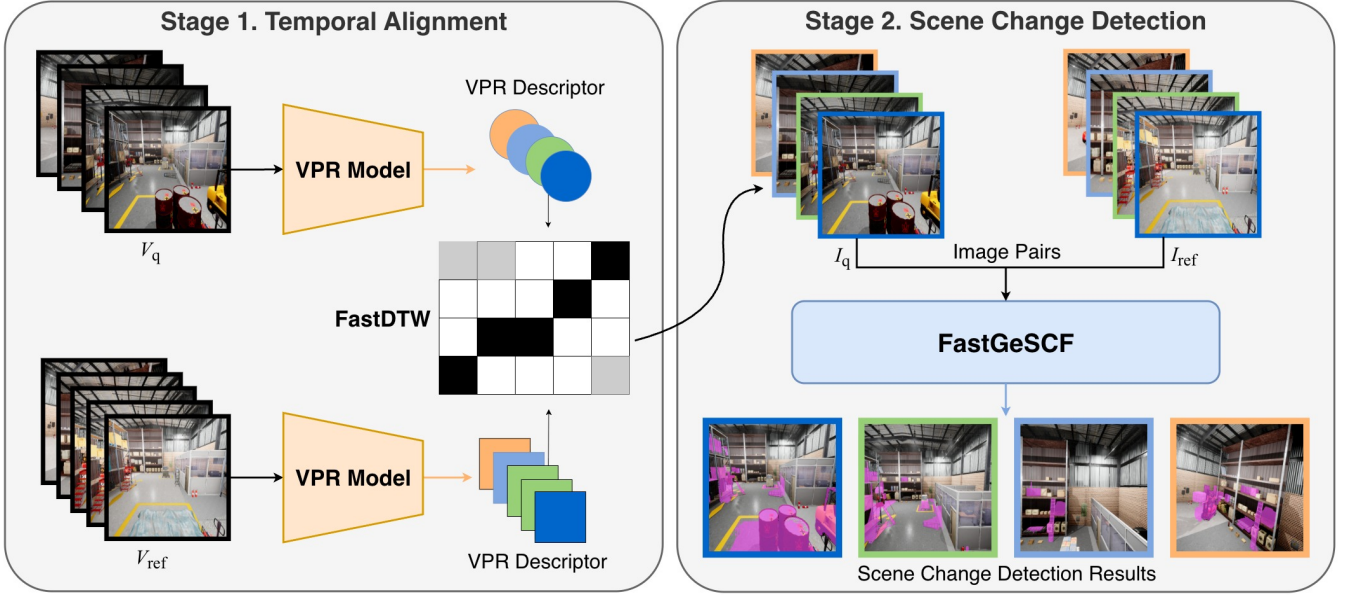


Fig. 1: **Proposed Two-Stage Pipeline.** We align unaligned videos using VPR and FastDTW (Stage 1), then detect changes using FastGeSCF (Stage 2), which employs a selective prompting strategy for efficiency.

**Abstract**—Scene change detection (SCD) is crucial for long-term robot autonomy but faces significant hurdles: the lack of temporal alignment in real-world video streams and the high latency of existing foundation model-based methods. To bridge the gap towards practical deployment, we propose a unified framework for unaligned videos captured during repeated traversals. First, we introduce a robust temporal alignment strategy integrating visual place recognition (VPR) with FastDTW, enabling reliable frame pairing even in visually degraded environments where standard retrieval fails. Second, we present FastGeSCF, a highly generalizable architecture that accelerates inference by replacing dense prompting with a novel coarse-to-fine selective prompting mechanism. By leveraging a cross-attention pseudo-mask generator and a grid point filter, we selectively prompt the Segment Anything Model (SAM) only at relevant regions. Experiments demonstrate that our method maintains competitive accuracy compared to computationally expensive baselines while reducing inference time by approximately  $7.9\times$  ( $\sim 0.88$  s/pair). This efficiency-accuracy balance makes it a viable solution for online robotic applications.

## I. INTRODUCTION

Scene change detection (SCD) is a critical capability for long-term autonomous robots operating in dynamic environments. It enables systems to identify significant alterations in their surroundings—such as moved objects, structural damage, or new obstacles—which is essential for applications ranging from disaster assessment [1] and warehouse monitoring [2] to urban patrol [3]. Unlike standard object detection, SCD requires comparing a current observation (query) against a pre-recorded map or database (reference) to pinpoint semantic differences while ignoring nuisance variations like lighting or weather changes.

Despite recent progress, deploying SCD on mobile robots faces two fundamental challenges: data alignment and inference efficiency. First, most existing SCD methods [4], [5] operate on the assumption of perfectly paired, pixel-aligned images. However, in real-world robotic missions (e.g., patrol or delivery), data is captured as continuous video streams where the robot follows a similar route but with varying speeds, trajectories, and sampling rates. This results in unaligned videos, where a simple one-to-one frame correspondence does not exist. While visual place recognition (VPR) methods [6], [7] have been used to retrieve reference frames, they typically treat images independently, ignoring

<sup>1</sup>Yi-Heng Yeh, Chuan-Yuan Huang, and Kuan-Wen Chen are with the Department of Computer Science, National Yang Ming Chiao Tung University, Hsinchu 300, Taiwan (e-mail of the corresponding author\* Kuan-Wen Chen: kuanwen@cs.nycu.edu.tw).

<sup>2</sup>Li-Yu Lu is a member of the Internet of Things Laboratory, Chunghwa Telecom Laboratories, Taoyuan, Taiwan (e-mail: uuiianlu318@cht.com.tw).

the sequential nature of robotic movement and leading to perceptual aliasing in repetitive environments.

Second, the trade-off between accuracy and speed remains a bottleneck. Recent state-of-the-art methods, such as GeSCF [8] and ZSSCD [9], have achieved impressive zero-shot generalization by leveraging foundation models like the Segment Anything Model (SAM) [10]. However, these approaches are computationally expensive. For instance, GeSCF requires prompting the heavy SAM decoder with a dense grid of points covering the entire image to generate masks, regardless of whether a change actually occurred. This redundancy results in high latency (e.g., several seconds per frame), rendering them impractical for online robotic applications that require rapid feedback.

To address these limitations, we present a practical framework designed for efficient scene change detection in unaligned videos. As shown in Fig. 1, our approach bridges the gap between robust sequence alignment and fast, generalizable change detection. For the alignment stage, we propose a strategy that combines robust VPR descriptors (SALAD [11]) with the FastDTW [12] algorithm, a linear-time approximation of dynamic time warping (DTW). Unlike simple frame retrieval, this formulation exploits the temporal consistency of video sequences, allowing us to robustly align query and reference frames even under severe visual degradation (e.g., dust or low light) where individual frame matching might fail.

For the detection stage, we introduce FastGeSCF, a novel architecture that significantly accelerates SAM-based SCD without compromising accuracy. We observe that in most SCD scenarios, changes are sparse. Therefore, decoding masks for the whole image is inefficient. FastGeSCF introduces a lightweight cross-attention pseudo-mask generator to rapidly identify potential change regions (appearance and disappearance) using SAM’s intermediate features. Based on these pseudo-masks, we design a grid point filter that selectively prompts the SAM decoder—which typically relies on a computationally expensive dense point grid—only at relevant locations. This selective prompting strategy eliminates redundant computations, achieving a speedup of approximately  $7.9\times$  compared to GeSCF.

In summary, our main contributions are as follows:

- We propose a practical temporal alignment strategy for SCD that integrates VPR with FastDTW. By leveraging the sequential nature of video data, our method significantly improves frame pairing recall in visually degraded environments compared to standard retrieval baselines.
- We present FastGeSCF, an efficient and domain-generalizable SCD framework. By introducing a cross-attention mechanism for accurate pseudo-mask generation and a grid point filtering strategy, we effectively solve the efficiency bottleneck of prior foundation model-based methods.
- Extensive experiments on the ChangeSim [2] and multiple real-world datasets demonstrate that our framework achieves competitive accuracy with a drastic reduction

in inference time (approximately 0.88 s/pair), paving the way for practical robotic deployment.

## II. RELATED WORK

### A. Visual Place Recognition and Temporal Alignment

VPR is fundamental for robot localization. Early methods aggregated local descriptors (e.g., VLAD [13], NetVLAD [6]), while recent approaches like SALAD [11] leverage foundation models (e.g., DINOv2 [14]) and optimal transport to achieve robustness against appearance changes. However, standard VPR typically treats queries independently, ignoring robotic motion continuity. To enforce temporal consistency, sequence matching techniques like SeqSLAM [15], [16] and DTW have been employed. While FastDTW [12] offers linear-time alignment for time-series, its integration with VPR for SCD in unaligned videos remains underexplored. In this work, we combine SALAD-based descriptors with FastDTW to robustly solve frame synchronization, addressing the alignment gap often overlooked in SCD literature that assumes perfectly paired inputs.

### B. Scene Change Detection

SCD aims to identify semantic changes while ignoring nuisance variations. Early deep learning approaches compared feature maps directly [17], [18] or used feature fusion strategies (e.g., Siamese networks) to categorize changes [4], [19]. Addressing spatial misalignment, SimSaC [7] proposed a detect-and-warp pipeline using optical flow, while RSCD [5] employed cross-attention and augmentation for viewpoint invariance. However, these supervised methods often require extensive domain-specific training and struggle to generalize to unseen environments (e.g., indoors vs. aerial), limiting their scalability for diverse robotic missions.

### C. Generalizable SCD with Foundation Models

To improve generalization, recent research leverages vision foundation models like SAM [10]. GeSCF [8] utilizes SAM’s internal feature correlations for zero-shot mask generation, while ZSSCD [9] combines SAM with tracking. Although achieving state-of-the-art accuracy, these methods incur high latency (e.g., seconds per frame) due to redundant operations like dense prompting [8], making them impractical for online robotics. FastGeSCF addresses this by introducing a lightweight cross-attention module to selectively prompt SAM only at relevant locations, achieving high efficiency without compromising the generalization power of foundation models.

## III. METHODOLOGY

In this section, we present the proposed framework for SCD in unaligned videos. As illustrated in Fig. 1, our pipeline consists of two main stages: (1) temporal alignment, which synchronizes the query video with the reference video to establish frame-level correspondences, and (2) FastGeSCF, a fast and generalizable SCD network that detects changes within these paired frames.

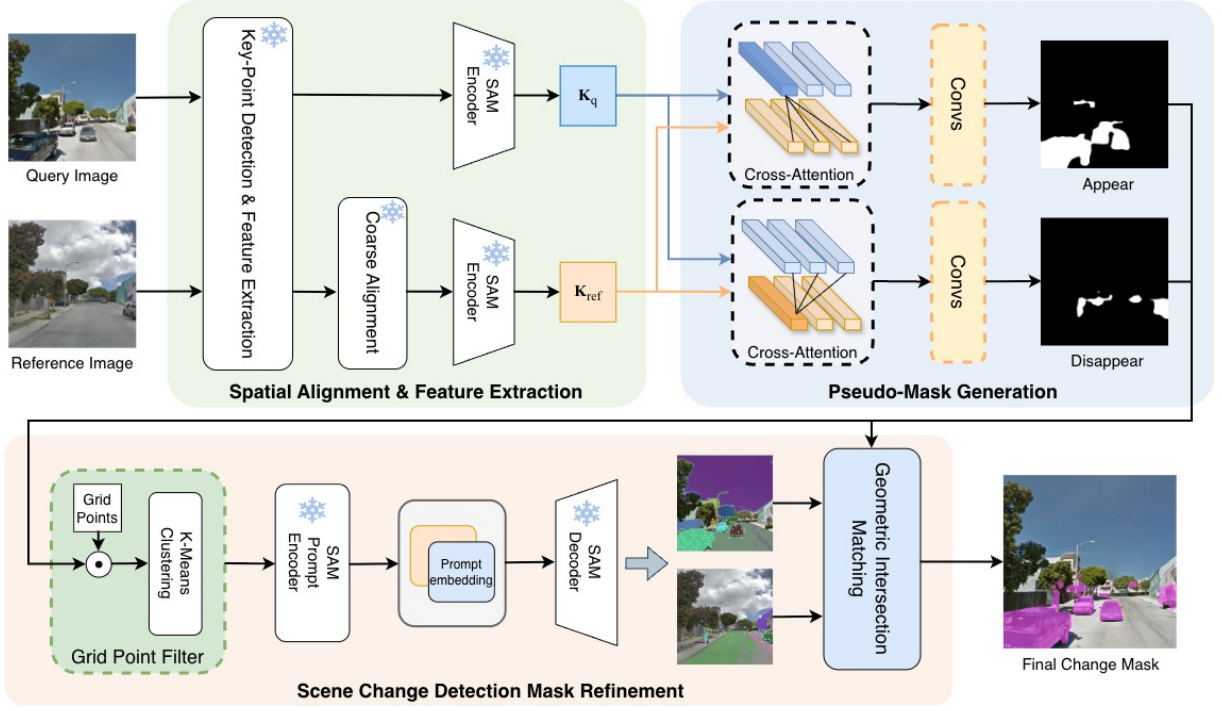


Fig. 2: **FastGeSCF Architecture.** The framework consists of (A) Spatial Alignment via LightGlue, (B) Pseudo-Mask Generation using cross-attention, and (C) Mask Refinement. A grid point filter selectively prompts SAM only at relevant regions to accelerate inference.

### A. Problem Formulation

Let  $V_q = \{I_q^1, I_q^2, \dots, I_q^M\}$  be a query video sequence capturing the current state of an environment, and  $V_{ref} = \{I_{ref}^1, I_{ref}^2, \dots, I_{ref}^N\}$  be a reference video sequence of the same environment captured at an earlier time. We assume that both sequences are collected by a robot traversing a spatially consistent route (i.e., repeated traversal), though with potentially different speeds and local viewpoints. Consequently,  $M \neq N$  and the temporal correspondence is non-linear but monotonic.

Our goal is twofold: first, to find a temporal mapping function  $\phi : \{1, \dots, M\} \rightarrow \{1, \dots, N\}$  that aligns each query frame  $I_q^t$  to its semantically corresponding reference frame  $I_{ref}^{\phi(t)}$ ; second, to generate a binary change mask  $M_{change}^t$  for each pair  $(I_q^t, I_{ref}^{\phi(t)})$ , where 1 indicates a scene change (e.g., appeared or disappeared objects) and 0 indicates unchanged regions.

### B. Stage 1: Robust Temporal Alignment

Existing SCD methods often assume perfectly paired images or rely on simple retrieval methods that ignore temporal continuity. To address this, we propose a robust alignment module combining VPR descriptors with sequence matching.

1) *Feature Extraction with SALAD:* We employ SALAD [11] as our VPR backbone. Unlike traditional global descriptors (e.g., NetVLAD [6]), SALAD leverages the DINOv2 [14] backbone and an optimal transport-based aggregation mechanism [20]. This allows it to produce robust global descriptors  $\mathbf{D} \in \mathbb{R}^d$  that are invariant to

significant viewpoint changes and illumination variations (e.g., day-to-night, dust), which are common in robotic applications.

2) *Sequence Matching via FastDTW:* Treating frame retrieval as independent queries often leads to perceptual aliasing (matching to a visually similar but geometrically distant location). Given the constraint that the robot traverses a similar path, the frame correspondence naturally exhibits temporal monotonicity. To leverage this property and enforce temporal consistency, we utilize the FastDTW algorithm [12].

While standard DTW finds the optimal alignment by minimizing the cumulative distance between two sequences of descriptors, its quadratic time and space complexity  $O(N^2)$  is prohibitive for long video sequences typical in robot navigation. FastDTW approximates the optimal path through a multi-level projection approach, reducing the complexity to linear time  $O(N)$ . This efficiency is crucial for the scalability of our pipeline. Formally, given the descriptor sequences  $\mathbf{D}_q$  and  $\mathbf{D}_{ref}$ , FastDTW computes the alignment path that minimizes the total cost, yielding a set of synchronized pairs  $\{(I_q^t, I_{ref}^{\tau_t})\}_{t=1}^M$ .

### C. Stage 2: FastGeSCF

After temporal alignment, the paired images may still exhibit spatial misalignments due to 6-degrees-of-freedom (6-DoF) pose differences. Furthermore, existing foundation model-based SCD methods (e.g., GeSCF [8]) suffer from high latency. To solve these issues, we propose FastGeSCF.

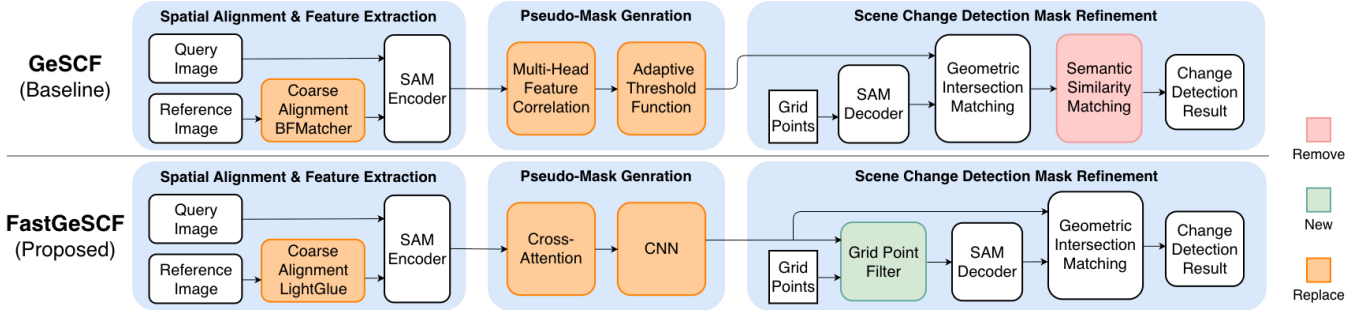


Fig. 3: **Structural Comparison.** GeSCF [8] (top) relies on dense prompting. FastGeSCF (bottom) streamlines the pipeline by removing redundancy, upgrading alignment, and adding a cross-attention prior for selective prompting.

Fig. 2 illustrates the overall architecture. We explicitly highlight the differences between our method and the baseline GeSCF in Fig. 3. The key improvements are:

- **Efficient Matching:** We replace the computationally heavy brute-force matcher with LightGlue [21] for faster and more accurate spatial alignment.
- **Context-Aware Prior:** We redesign the pseudo-mask generator using a cross-attention mechanism to capture semantic context, overcoming the limitations of simple feature correlation.
- **Selective Prompting:** We introduce a grid point filter to selectively prompt SAM, avoiding redundant full-image segmentation.

The framework comprises three substages: Spatial Alignment and Feature Extraction, Pseudo-Mask Generation, and Mask Refinement.

1) *Spatial Alignment and Feature Extraction:* Although temporally aligned, the image pair  $(I_q, I_{ref})$  requires pixel-level registration. We adopt a feature-based registration approach.

First, we extract local keypoints and descriptors using SuperPoint [22]. For feature matching, we adopt LightGlue [21]. Unlike simple heuristic matchers, LightGlue employs a deep attention mechanism to reason about the assignment of points, providing superior robustness to viewpoint changes while remaining highly efficient on the GPU. Based on the matched keypoints, we estimate a homography matrix and warp the reference image  $I_{ref}$  to align with the query image  $I_q$ . A visibility mask is generated to ignore regions that fall out of the field of view after warping.

Once spatially aligned, we extract high-level semantic features for change detection. Following GeSCF [8], we utilize the frozen image encoder of SAM [10]. Specifically, we extract the key embeddings ( $\mathbf{K}$ ) from the multi-head self-attention layers of the encoder. As noted in [23], key features in Vision Transformers capture robust semantic correspondences suitable for matching. Let  $\mathbf{K}_q$  and  $\mathbf{K}_{ref}$  denote the extracted feature maps for the query and aligned reference images, respectively, where  $\mathbf{K} \in \mathbb{R}^{H' \times W' \times C}$ . These features serve as the input for the subsequent pseudo-mask generation.

2) *Cross-Attention Pseudo-Mask Generation:* After feature extraction, we obtain the semantic key embeddings

$\mathbf{K}_q$  and  $\mathbf{K}_{ref}$  from the query and reference images. While GeSCF [8] computes a simple cosine similarity map between these features, we observe that this approach treats object appearance and disappearance symmetrically and often fails to capture context, leading to noisy initial estimations. To address this, we introduce a directed cross-attention module that explicitly models the relationship between the two views.

We adopt a standard attention mechanism [24] to explicitly model the semantic discrepancies. Let  $\mathbf{X}, \mathbf{Y} \in \mathbb{R}^{H' \times W' \times C}$  be two feature maps. The cross-attention operation  $CA(\mathbf{X}, \mathbf{Y})$  queries  $\mathbf{X}$  against the context  $\mathbf{Y}$ , effectively highlighting regions in  $\mathbf{X}$  that cannot be explained by  $\mathbf{Y}$ . Specifically, the features are projected into query ( $\mathbf{Q}_x$ ), key ( $\mathbf{K}_y$ ), and value ( $\mathbf{V}_y$ ) embeddings via linear projections:

$$CA(\mathbf{X}, \mathbf{Y}) = \text{softmax} \left( \frac{\mathbf{Q}_x(\mathbf{K}_y)^T}{\sqrt{d_k}} \right) \mathbf{V}_y, \quad (1)$$

where  $d_k$  is the key dimension. We employ a bidirectional strategy to capture both appearance and disappearance:

$$\hat{M}_{\text{appear}} = \mathcal{F}(CA(\mathbf{K}_q, \mathbf{K}_{ref})), \quad (2)$$

$$\hat{M}_{\text{disappear}} = \mathcal{F}(CA(\mathbf{K}_{ref}, \mathbf{K}_q)), \quad (3)$$

where  $\mathcal{F}$  comprises a  $3 \times 3$  convolution followed by ReLU, a  $1 \times 1$  two-class head followed by ReLU, and bilinear upsampling.  $\hat{M}_{\text{appear}}$  highlights new objects, while  $\hat{M}_{\text{disappear}}$  identifies missing ones. This explicit modeling provides a comprehensive prior for the refinement stage.

3) *Mask Refinement via Grid Point Filtering:* The primary bottleneck in existing SAM-based SCD methods [8], [9] is the heavy reliance on the mask decoder. Unlike our approach, methods like GeSCF perform dense prompting, requiring the model to process a grid of points (e.g.,  $32 \times 32 = 1024$  points) to segment the entire image regardless of actual changes, which is computationally prohibitive for video processing. However, in the context of SCD, we are only interested in the changed regions. Leveraging the pseudo-masks derived in the previous step (Sec. III-C.2), we propose a grid point filter strategy composed of two steps to drastically reduce the number of prompts.

1) **Background Elimination.** Let  $G = \{p_1, p_2, \dots, p_N\}$  be the standard uniform grid of points defined over the image spatial dimensions. Instead of prompting SAM with the

TABLE I: **Temporal Alignment Performance on ChangeSim.** We use the Normal sequence as the query and align it against Dark/Dust reference sequences. Results show Recall@K accuracy (success if error < 1 m).

| Method               | Query  | Ref  | R@1           | R@5           | R@10          |
|----------------------|--------|------|---------------|---------------|---------------|
| VPR Only             | Normal | Dark | 0.3443        | 0.4950        | <b>0.5627</b> |
|                      | Normal | Dust | 0.4955        | 0.6244        | 0.6839        |
| <b>VPR + FastDTW</b> | Normal | Dark | <b>0.4592</b> | <b>0.4974</b> | 0.5464        |
|                      | Normal | Dust | <b>0.6164</b> | <b>0.6585</b> | <b>0.6905</b> |

entire set  $G$ , we filter out points that fall into the background regions of the pseudo-masks. For each direction (appearance and disappearance), the set of valid candidate points  $P_{\text{cand}}$  is defined as:

$$P_{\text{cand}} = \{p \in G \mid \hat{M}(p) = 1\}, \quad (4)$$

where  $\hat{M} \in \{\hat{M}_{\text{appear}}, \hat{M}_{\text{disappear}}\}$ . This step typically removes over 90% of the grid points in static scenes, directly translating to inference speedup.

**2) Redundancy Reduction via Clustering.** Even within the changed regions, adjacent grid points often correspond to the same object instance, leading to redundant mask generation. To further optimize efficiency, we apply K-means clustering [25] to  $P_{\text{cand}}$  to obtain a sparse set of representative prompts  $P_{\text{prompt}}$ . The number of clusters is dynamically set to  $k = \max(1, \lfloor r_s \cdot |P_{\text{cand}}| \rfloor)$ , where  $r_s$  is a sampling ratio set to 0.5. This ensures that the number of prompts scales adaptively with the complexity of the scene change.

**Geometric Intersection Matching.** Finally, the selected prompts  $P_{\text{prompt}}$  are encoded via SAM’s prompt embedding layer and fed into the mask decoder to generate high-quality segmentation masks. Since SAM is class-agnostic, it might segment a larger object than the actual change (e.g., the entire building instead of just new graffiti). To filter these false positives, we employ Geometric Intersection Matching (GIM) adapted from GeSCF [8]. A generated mask  $M_{\text{SAM}}$  is accepted as a valid change only if it has sufficient overlap with the initial pseudo-mask:

$$\frac{|M_{\text{SAM}} \cap \hat{M}|}{|M_{\text{SAM}}|} > \theta, \quad (5)$$

where we set  $\theta = 0.5$ . The final change mask is the union of all accepted masks from both streams.

By replacing the dense grid query with our targeted prompting strategy, FastGeSCF avoids wasteful computation on unchanged background areas, achieving the significant speedup reported in our experiments.

#### IV. EXPERIMENTAL SETUP

##### A. Datasets and Metrics

We train FastGeSCF on the VL-CMU-CD [17] training set. To evaluate cross-domain generalization and robustness, we test on diverse datasets without fine-tuning:

- ChangeSim [2]: A photorealistic synthetic dataset used to evaluate temporal alignment under Normal, Dark, and

TABLE II: **Impact of Alignment Strategy on SCD Performance.** SCD F1-scores on ChangeSim using FastGeSCF with ground truth (GT) versus VPR-based pairing.

| Pairing Strategy            | Normal        | Dust          | Dark          |
|-----------------------------|---------------|---------------|---------------|
| GT Pair (Upper Bound)       | <b>0.5400</b> | 0.4790        | 0.4010        |
| VPR Only                    | 0.4900        | 0.4866        | 0.4063        |
| <b>VPR + FastDTW (Ours)</b> | 0.5030        | <b>0.4900</b> | <b>0.4133</b> |

Dust conditions. We report Recall@1 (success if error < 1 m [7]).

- SCD Benchmarks: We evaluate SCD performance on the VL-CMU-CD test set, PSCD [26], and the ChangeVPR benchmark [8] (which includes St Lucia [27], SF-XL [28], and Nordland [16]). We report the F1-score of the change class.
- Oxford RobotCar [29]: Used for qualitative analysis of real-world unaligned videos.

##### B. Baselines

For alignment, we compare our method against a standard VPR-only approach (frame-by-frame retrieval without temporal constraints, following the retrieval strategy in SimSaC [7]) and ground truth (GT) pairing. For SCD, we compare against state-of-the-art generalizable methods: GeSCF [8] and ZSSCD [9].

##### C. Implementation Details

Our framework is implemented in PyTorch [30] on a single NVIDIA RTX 3090 GPU (24GB). **Alignment:** We use the pre-trained SALAD [11] model (with a Vision Transformer, ViT-L/14 backbone) to extract 8448-dim global descriptors. FastDTW is applied with a radius of 1, ensuring linear time complexity  $O(N)$  while maintaining sufficient local alignment accuracy. **FastGeSCF:** We use a frozen SAM ViT-H encoder [10]; the added cross-attention learns Q/K/V projections, and only it and  $\mathcal{F}$  are trained. The only augmentation is a joint horizontal flip of each image pair and GT mask (probability 0.5). We train for 50 epochs with Adam ( $lr = 10^{-4}$ ), cosine annealing, and batch size 4. The pseudo-mask generator is supervised directly by the full VL-CMU-CD GT masks using pixel-wise weighted cross-entropy (0.025 unchanged, 0.975 change) [5]. For efficiency evaluation, we measure the average inference time per image pair, encompassing feature extraction, matching, and mask generation.

#### V. RESULTS AND DISCUSSION

In this section, we evaluate the proposed framework from two perspectives: the robustness of the temporal alignment in challenging environments and the accuracy of scene change detection across diverse domains.

##### A. Robustness of Temporal Alignment

We first validate the effectiveness of our temporal alignment module (Stage 1) on the ChangeSim dataset [2].



TABLE III: **Quantitative Comparison on SCD Benchmarks.** We report the F1-score for accuracy and average inference time per image pair. Our method achieves comparable accuracy to state-of-the-art but is significantly faster.

| Method                  | F1-score $\uparrow$ |              |              |              |              | Avg F1       | Time (s) $\downarrow$ |
|-------------------------|---------------------|--------------|--------------|--------------|--------------|--------------|-----------------------|
|                         | VL-CMU-CD (Test)    | PSCD         | Nordland     | St Lucia     | SF-XL        |              |                       |
| GeSCF [8]               | 0.754               | <b>0.540</b> | <b>0.590</b> | 0.621        | 0.712        | 0.643        | 6.982                 |
| ZSSCD [9]               | 0.516               | 0.440        | 0.230        | 0.470        | 0.496        | 0.430        | 10.128                |
| <b>FastGeSCF (Ours)</b> | <b>0.760</b>        | 0.526        | 0.542        | <b>0.726</b> | <b>0.736</b> | <b>0.658</b> | <b>0.888</b>          |

This dataset is particularly suitable for this evaluation as it provides ground-truth 6-DoF poses, allowing us to quantify retrieval accuracy under varying environmental conditions. Specifically, we use the Normal sequence as the query and align it against reference sequences captured under Dark (low illumination) and Dust (visual obstruction) conditions.

Table I compares the performance of a standard VPR-only approach (retrieving frames independently) against our proposed VPR + FastDTW strategy. Following [7], a retrieval is considered correct if the retrieved frame is within 1 meter of the query frame.

**Impact of Sequence Matching:** As shown in Table I, incorporating FastDTW significantly enhances alignment accuracy, particularly in visually degraded environments. While the VPR-only baseline achieves reasonable performance, it suffers from perceptual aliasing—matching frames that look similar locally but are geometrically distant. By enforcing temporal consistency constraints, our method filters out these outliers. Notably, in the Dark scenario, our method achieves an absolute improvement of 11.5% in Recall@1 (from 0.3443 to 0.4592). Similarly, in the Dust scenario, we observe a 12.1% increase (from 0.4955 to 0.6164). Since the subsequent SCD stage relies heavily on the quality of the top-1 retrieved pair, this substantial boost is critical for the overall system performance. These results demonstrate that leveraging the sequential nature of video data is essential for robust robot perception in real-world, imperfect conditions.

### B. Scene Change Detection Accuracy

Next, we evaluate the SCD performance. We analyze how different pairing strategies affect detection quality and compare our FastGeSCF model against state-of-the-art methods on unseen domains.

*1) Impact of Alignment on SCD:* To understand how temporal alignment quality propagates to the final detection, we conducted a comparative analysis on the ChangeSim dataset using the FastGeSCF model. We compared three pairing strategies: (1) **GT Pair:** Pairs determined by ground-truth pose distance; (2) **VPR:** Pairs from raw VPR retrieval; and (3) **VPR + FastDTW:** Pairs from our proposed pipeline.

The results in Table II demonstrate the effectiveness of our alignment strategy. Under the Normal condition, our method (0.503) successfully bridges the performance gap between the raw VPR baseline (0.490) and the GT upper bound (0.540).

More importantly, under visually degraded conditions (Dust and Dark), our automatically derived pairs yield F1-

scores comparable to, and in some cases slightly exceeding, those obtained with GT pairs (e.g., 0.413 vs. 0.401 in Dark). This result is counterintuitive but explicable: GT pairs are selected based on strict physical distance, which may force a match with a heavily occluded or uninformative reference frame. In contrast, our VPR+FastDTW pipeline explicitly optimizes for visual similarity within temporal constraints. This suggests that our pipeline produces pairs suitable for SCD that are robust for visual tasks even when precise physical localization is challenged.

*2) SCD Performance and Generalization:* We evaluate FastGeSCF on the source domain (VL-CMU-CD test set) and four unseen datasets (PSCD, Nordland, St Lucia, SF-XL) to assess both learning capability and generalization. Table III summarizes the quantitative results. FastGeSCF achieves the highest average F1-score of 0.658 across all benchmarks, outperforming GeSCF (0.643) and ZSSCD (0.430). Specifically, our method excels in the source domain (0.760) and demonstrates strong transferability on unseen datasets like St Lucia (0.726) and SF-XL (0.736). The improvement over GeSCF validates that our cross-attention mechanism (Sec. III-C.2) generates higher-quality priors than simple feature correlation. This semantic robustness allows the subsequent refinement stage to focus strictly on relevant changes, maintaining high accuracy without the need for dense sampling. Crucially, as discussed in the next section, FastGeSCF achieves this competitive accuracy with a fraction of the computational cost, making it the only viable option for real-time robotic applications among the compared methods.

### C. Runtime and Efficiency Analysis

High computational cost is a primary barrier for foundation model-based SCD in robotics. We analyze the efficiency of FastGeSCF to demonstrate its suitability for online applications.

Table III (last column) presents the average inference time per image pair. FastGeSCF achieves an inference speed of 0.888 seconds, which is approximately  $7.9\times$  faster than GeSCF (6.982s) and  $11.4\times$  faster than ZSSCD (10.128s). Regarding the temporal alignment (Stage 1), since it operates on low-dimensional global descriptors ( $1 \times 8448$ ) rather than pixel-level processing, the computational overhead is marginal compared to the mask generation in Stage 2. Consequently, our full pipeline transforms SCD from an offline post-processing step to a near-real-time operation suitable for mobile robots.

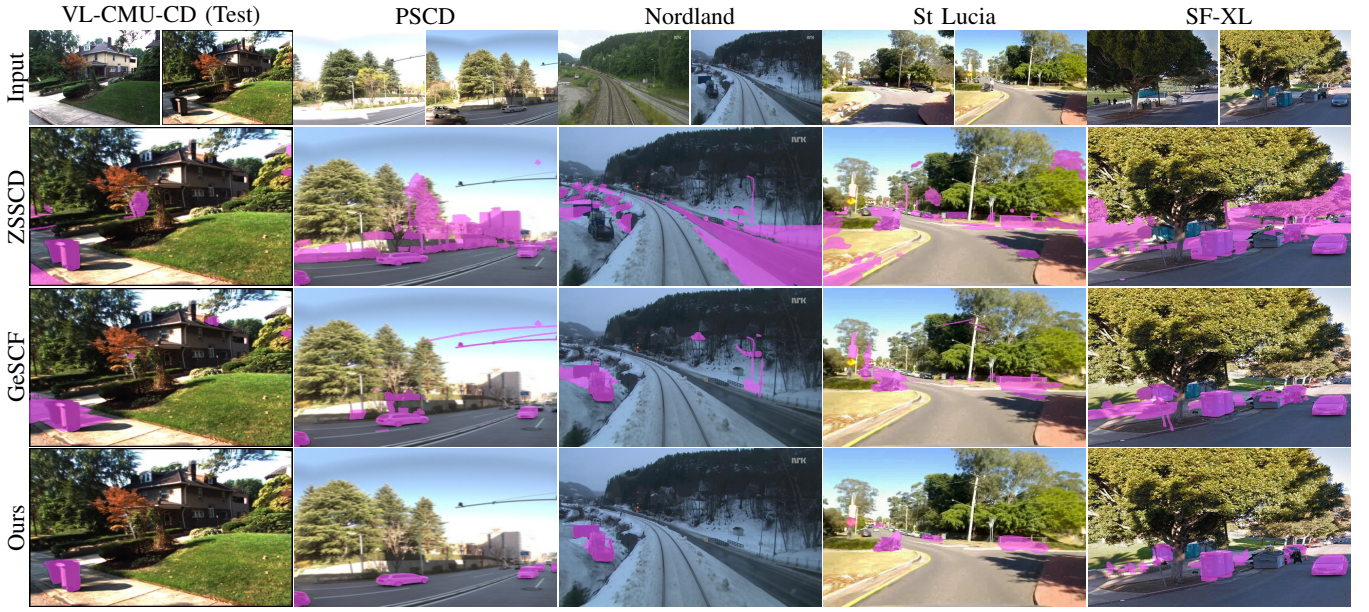


Fig. 4: **Qualitative Results.** Comparison on five datasets. FastGeSCF (bottom) robustly detects changes while ignoring environmental noise (e.g., shadows in St Lucia), outperforming ZSSCD and GeSCF.

TABLE IV: **Component-wise Runtime Analysis.** Average latency per image pair across all datasets. Total includes spatial alignment and feature extraction. GP: Grid Point.

| Method Configuration    | Pseudo-Mask (s) | Refinement (s) | Total Latency (s) |
|-------------------------|-----------------|----------------|-------------------|
| GeSCF (Baseline)        | 0.030           | 5.653          | 6.982             |
| GeSCF + GP Filter       | 0.030           | 2.820          | 4.149             |
| <b>FastGeSCF (Ours)</b> | 0.176           | <b>0.529</b>   | <b>0.888</b>      |

#### D. Ablation Studies

We conduct ablation studies to validate the design choices, specifically analyzing how component quality directly impacts system latency.

1) *Efficacy of Cross-Attention on Efficiency:* A key hypothesis of our framework is that investing computation in a better initial prior reduces the burden on the heavy SAM decoder. To validate this, we analyze the component-wise latency in Table IV.

We compare our full pipeline against a baseline, GeSCF + GP Filter, where our grid point filter is applied directly to GeSCF’s correlation-based pseudo-masks. As shown in Table IV, although GeSCF’s simple correlation map is faster to compute (0.03s) than our cross-attention module (0.176s), it produces a noisy prior. This noise forces the filter to retain many redundant grid points to ensure recall, resulting in a high refinement latency of 2.82s. In contrast, our cross-attention module generates a significantly more precise prior. This precision allows the grid point filter to discard over 90% of irrelevant prompts, drastically reducing the dominant refinement stage to just 0.529s. This confirms a critical design insight: a slightly costlier but accurate pseudo-mask yields a net positive in total system latency by avoiding wasteful segmentation of static background regions.

TABLE V: **Ablation on Grid Point Downsampling Strategies.** We adopt the proportional strategy ( $0.5 \times |P_{\text{cand}}|$ ) for the best trade-off.

| Strategy            | # Clusters ( $k$ )              | F1-score     | Time (s)     |
|---------------------|---------------------------------|--------------|--------------|
| None                | All candidates                  | <b>0.661</b> | 1.196        |
| Fixed               | $k = 100$                       | 0.660        | 0.978        |
| Fixed               | $k = 50$                        | 0.655        | 0.838        |
| <b>Proportional</b> | $0.25 \times  P_{\text{cand}} $ | 0.646        | <b>0.770</b> |
| <b>Proportional</b> | $0.5 \times  P_{\text{cand}} $  | 0.658        | 0.888        |

2) *Impact of K-Means Downsampling Strategy:* In the refinement stage, we employ K-means clustering to reduce redundant prompts within changed regions. Table V investigates the trade-off between the number of clusters and performance. Using a fixed number of clusters (e.g., 50 or 100) lacks adaptability; it may under-sample large changes or over-sample small ones. Conversely, our proportional strategy ( $k = r_s \cdot |P_{\text{cand}}|$ ) dynamically adapts to the scene complexity. The results show that  $r_s = 0.5$  offers the optimal balance, achieving an F1-score (0.658) comparable to the baseline without downsampling (0.661) but with significantly lower latency (0.888s vs. 1.196s). Aggressive downsampling ( $r_s = 0.25$ ) yields further speedups but begins to degrade detection accuracy (0.646).

#### E. Qualitative Results

We provide visual comparisons to demonstrate the robustness of FastGeSCF in real-world scenarios.

1) *Robustness to Environmental Variations:* Fig. 4 demonstrates robustness. In St Lucia and SF-XL, strong shadows confuse GeSCF, causing false positives. FastGeSCF’s cross-attention module effectively contextualizes query features against the reference, ignoring illumination

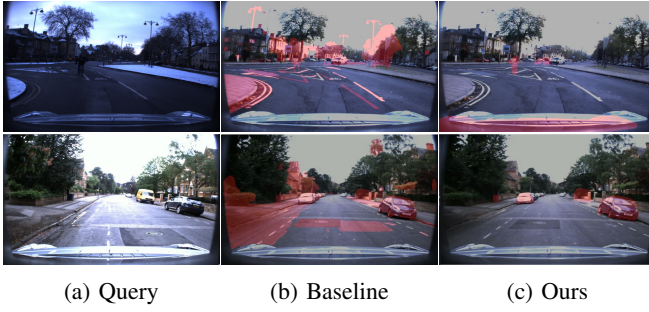


Fig. 5: **Real-World Unaligned Video (Oxford RobotCar).** (a) Query. (b) Baseline (VPR+GeSCF) fails due to misalignment. (c) Ours (VPR+FastDTW+FastGeSCF) robustly detects changes despite viewpoint shifts. See **supplementary video** for full sequences.

inconsistencies to focus on physical changes (e.g., cars).

2) *Real-World Unaligned Video*: We evaluate the full pipeline on the Oxford RobotCar dataset [29] (see Fig. 5). Video: [https://youtu.be/d1J3pL\\_wxRw](https://youtu.be/d1J3pL_wxRw). While the baseline (VPR+GeSCF) struggles with perspective shifts, our approach mitigates misalignment artifacts by effectively synchronizing frames via FastDTW and LightGlue. This allows FastGeSCF to focus on genuine semantic changes rather than registration errors, highlighting its potential for autonomous driving deployment.

## VI. CONCLUSION

We presented a practical framework for SCD in unaligned videos, addressing the dual challenges of data alignment and inference efficiency. By integrating VPR with FastDTW, we achieve robust temporal alignment superior to standard retrieval, particularly under challenging visibility. Furthermore, our proposed FastGeSCF resolves the computational bottleneck of SAM-based methods. Through a novel cross-attention and filtering mechanism, we achieve a 7.9× speedup over existing methods while maintaining competitive average accuracy across diverse benchmarks. This work effectively bridges the gap between high-performance foundation models and the practical latency constraints of robotic applications, making SCD tractable for online deployment. Future work will focus on end-to-end optimization and deployment on edge devices.

## ACKNOWLEDGMENT

This work was supported in part by the National Science and Technology Council, Taiwan, under Grant 114-2221-E-A49-054-MY3, Grant 113-2221-E-A49-164-MY3, and Grant 115-2634-F-A49-014.

## REFERENCES

- [1] K. Ahn, S. Han, S. Park, *et al.*, “Generalizable disaster damage assessment via change detection with vision foundation model,” in *Proc. AAAI*, 2025, pp. 27 784–27 792.
- [2] J.-M. Park, J.-H. Jang, S.-M. Yoo, *et al.*, “Changesim: Towards end-to-end online scene change detection in industrial indoor environments,” in *Proc. IROS*, 2021, pp. 8578–8585.
- [3] T. Alpherts, S. Ghebreab, and N. van Noord, “Emplace: Self-supervised urban scene change detection,” in *Proc. AAAI*, 2025, pp. 1737–1745.
- [4] G.-H. Wang, B.-B. Gao, and C. Wang, “How to reduce change detection to semantic segmentation,” *Pattern Recognition*, vol. 138, p. 109384, 2023.
- [5] C.-J. Lin, S. Garg, T.-J. Chin, and F. Dayoub, “Robust scene change detection using visual foundation models and cross-attention mechanisms,” in *Proc. ICRA*, 2025, pp. 8337–8343.
- [6] R. Arandjelovic, P. Gronat, A. Torii, *et al.*, “Netvlad: Cnn architecture for weakly supervised place recognition,” in *Proc. CVPR*, 2016, pp. 5297–5307.
- [7] J.-M. Park *et al.*, “Dual task learning by leveraging both dense correspondence and mis-correspondence for robust change detection with imperfect matches,” in *Proc. CVPR*, 2022, pp. 13 749–13 759.
- [8] J.-W. Kim and U.-H. Kim, “Towards generalizable scene change detection,” in *Proc. CVPR*, 2025, pp. 24 463–24 473.
- [9] K. Cho, D. Y. Kim, and E. Kim, “Zero-shot scene change detection,” in *Proc. AAAI*, 2025, pp. 2509–2517.
- [10] A. Kirillov, E. Mintun, N. Ravi, *et al.*, “Segment anything,” in *Proc. ICCV*, 2023, pp. 4015–4026.
- [11] S. Izquierdo and J. Civera, “Optimal transport aggregation for visual place recognition,” in *Proc. CVPR*, 2024, pp. 17 658–17 668.
- [12] S. Salvador and P. Chan, “Toward accurate dynamic time warping in linear time and space,” *Intelligent Data Analysis*, vol. 11, no. 5, pp. 561–580, 2007.
- [13] H. Jégou, M. Douze, C. Schmid, and P. Pérez, “Aggregating local descriptors into a compact image representation,” in *Proc. CVPR*, 2010, pp. 3304–3311.
- [14] M. Oquab, T. Darcet, T. Moutakanni, *et al.*, “Dinov2: Learning robust visual features without supervision,” *arXiv:2304.07193*, 2023.
- [15] M. J. Milford and G. F. Wyeth, “Seqslam: Visual route-based navigation for sunny summer days and stormy winter nights,” in *Proc. ICRA*, 2012, pp. 1643–1649.
- [16] N. Sünderhauf, P. Neubert, and P. Protzel, “Are we there yet? challenging seqslam on a 3000 km journey across all four seasons,” in *Proc. ICRA*, 2013.
- [17] P. F. Alcantarilla, S. Stent, G. Ros, *et al.*, “Street-view change detection with deconvolutional networks,” *Autonomous Robots*, vol. 42, pp. 1301–1322, 2018.
- [18] E. Guo, X. Fu, J. Zhu, *et al.*, “Learning to measure change: Fully convolutional siamese metric networks for scene change detection,” *arXiv:1810.09111*, 2018.
- [19] R. C. Daudt, B. Le Saux, and A. Boulch, “Fully convolutional siamese networks for change detection,” in *Proc. ICIP*, 2018, pp. 4063–4067.
- [20] M. Cuturi, “Sinkhorn distances: Lightspeed computation of optimal transport,” *NeurIPS*, vol. 26, 2013.
- [21] P. Lindenberger, P.-E. Sarlin, and M. Pollefeys, “Lightglue: Local feature matching at light speed,” in *Proc. ICCV*, 2023, pp. 17 627–17 638.
- [22] D. DeTone, T. Malisiewicz, and A. Rabinovich, “Superpoint: Self-supervised interest point detection and description,” in *Proc. CVPRW*, 2018, pp. 224–236.
- [23] S. Amir, Y. Gandelsman, S. Bagon, and T. Dekel, “Deep vit features as dense visual descriptors,” *arXiv:2112.05814*, 2021.
- [24] A. Vaswani, N. Shazeer, N. Parmar, *et al.*, “Attention is all you need,” *NeurIPS*, vol. 30, 2017.
- [25] J. B. McQueen, “Some methods of classification and analysis of multivariate observations,” in *Proc. 5th Berkeley Symp. Math. Stat. Prob.*, 1967, pp. 281–297.
- [26] K. Sakurada, M. Shibuya, and W. Wang, “Weakly supervised silhouette-based semantic scene change detection,” in *Proc. ICRA*, 2020, pp. 6861–6867.
- [27] M. J. Milford and G. F. Wyeth, “Mapping a suburb with a single camera using a biologically inspired slam system,” *IEEE Trans. Robot.*, vol. 24, no. 5, pp. 1038–1053, 2008.
- [28] G. Berton, C. Masone, and B. Caputo, “Rethinking visual geo-localization for large-scale applications,” in *Proc. CVPR*, 2022, pp. 4878–4888.
- [29] W. Maddern *et al.*, “1 year, 1000 km: The oxford robotcar dataset,” *Int. J. Robot. Res.*, vol. 36, no. 1, pp. 3–15, 2017.
- [30] A. Paszke *et al.*, “Pytorch: An imperative style, high-performance deep learning library,” in *NeurIPS*, 2019, vol. 32.